

CYBERSECURITY AWARENESS WORKSHEET

Spot the Secure Website: HTTP vs HTTPS

Worksheet completed by Selina

This activity helps participants identify secure vs insecure websites by analysing URL structure, browser indicators, and trust signals.

The goal is to improve awareness of how to recognise potentially unsafe websites and understand that HTTPS alone does not guarantee legitimacy.

Section 1: Website Security Evaluation

Website	URL	Indicators	Secure? (Yes/No)
A	http://online-banking-login.example.com	No padlock, HTTP	No
B	https://university.edu/login	Padlock, HTTPS	Yes
C	https://secure-paypal-login.example.net	HTTPS but suspicious domain	No

Questions:

Which website is most secure? Why?

B, as it has most reliable indicators.

Why is HTTP considered unsafe?

Doesn't encrypt communication.

Does HTTPS always mean a site is trustworthy? Explain.

Although it provides encrypted communication, the website itself can be unsafe.

Section 2: Safe vs Unsafe Links

Evaluate the following links-

Task:

- Mark each link as Safe / Suspicious
- Identify one red flag for each suspicious link

1. <https://paypal.com/login>

Yes, as it is a common website.

2. <https://paypal-secure-login.verify-now.example.com>

No, as it looks too long.

3. <https://uwa.edu.au>

Yes, as it looks standard.

4. <http://free-prizes.example.net>

No, as free prizes seems unreliable.

Section 3: File Safety (Malware Awareness)

Determine whether these files are safe:

- invoice.pdf.exe
- report.docx
- update_patch_v2.zip
- photo.png

Questions:

- Which file is most suspicious? Why?
- What is a common trick used in malicious file names?

Invoice.pdf.exe – suspicious, as it double extension.

Report.docx – safe, as it is a normal word document.

Update_patch_v2.zip – Suspicious, as it could contain threatening code.

Photo.png – Safe, as it is a standard photo format.

Section 4: Password Strength Analysis

Rank the following passwords from weakest → strongest:

- ☐1 123456
- ☐2 password2024
- ☐3 sam12345
- ☐4 UWA!secure9
- ☐5 Xy9#kL2pQ

Questions:

- What makes a password strong?
- Why is password reuse dangerous?

Length, randomness, numbers, special characters, capital letters, etc.

If an attacker gets one passwords, then they can compromise other sites as well.

Section 5: Social Engineering Scenarios

Scenario 1:

You receive a message from “IT Support” asking for your password to fix an issue.

Scenario 2:

A friend messages you urgently asking for money.

Questions:

- What are the warning signs?
- What would you do in each situation?

Scenario 1 – Asking for your password (warning signs), ignore (action)

Scenario 2 – Asking for money (warning signs), call the person (action)

Section 6: Quick Knowledge Check (Mixed Questions)

Multiple Choice:

1. What does HTTPS provide?
 - ☒ Encryption
 - ☐ Faster internet speed
 - ☐ Virus protection
2. What is phishing?
 - ☐ A type of malware
 - ☒ A social engineering attack
 - ☐ A firewall

True / False:

- HTTPS guarantees a website is completely safe (T / F) – False
- Strong passwords should include symbols and numbers (T / F) – True

Section 4: Reflection

What new cybersecurity concept did you learn?

HTTPS provides encryption for safe communication, but doesn't necessarily mean the website is safe.

Which section did you find most difficult?

HTTPS section.

How will this activity change your online behaviour?

Better inspect website links.
